Heartbeat Detection in Multi-site Wearable PPG

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Train / test split: by subject, 80 / 20

Multisite-PPG

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WF-PPG

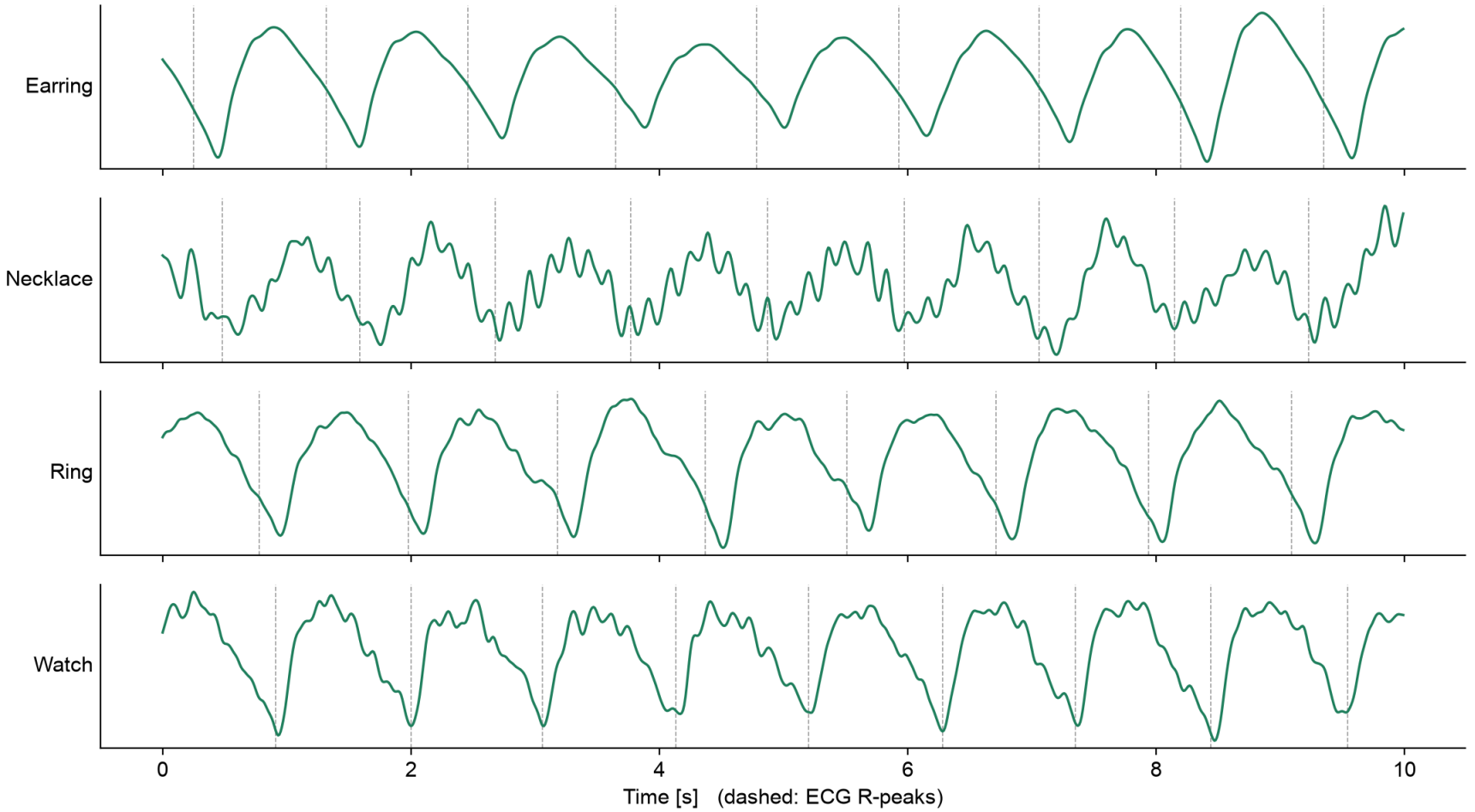
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Wrist PPG During Exercise

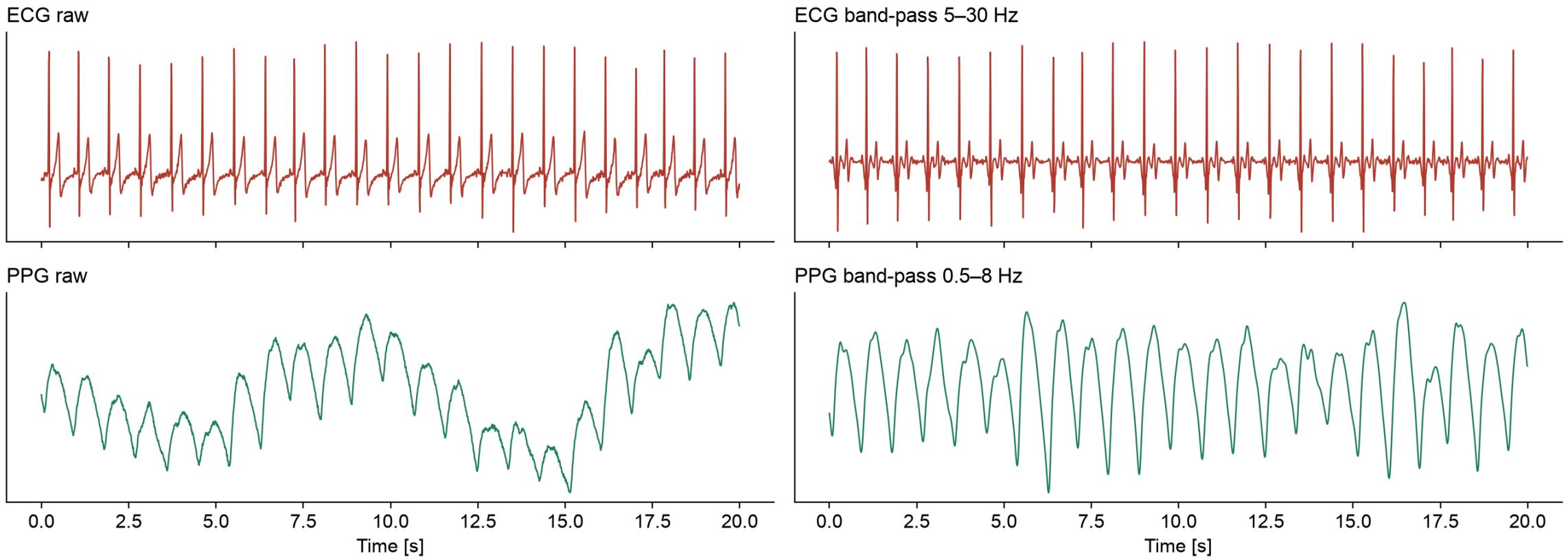
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Split by subject: no person appears in both sets, so the test measures generalization to new people.

Same heart, four very different waveforms

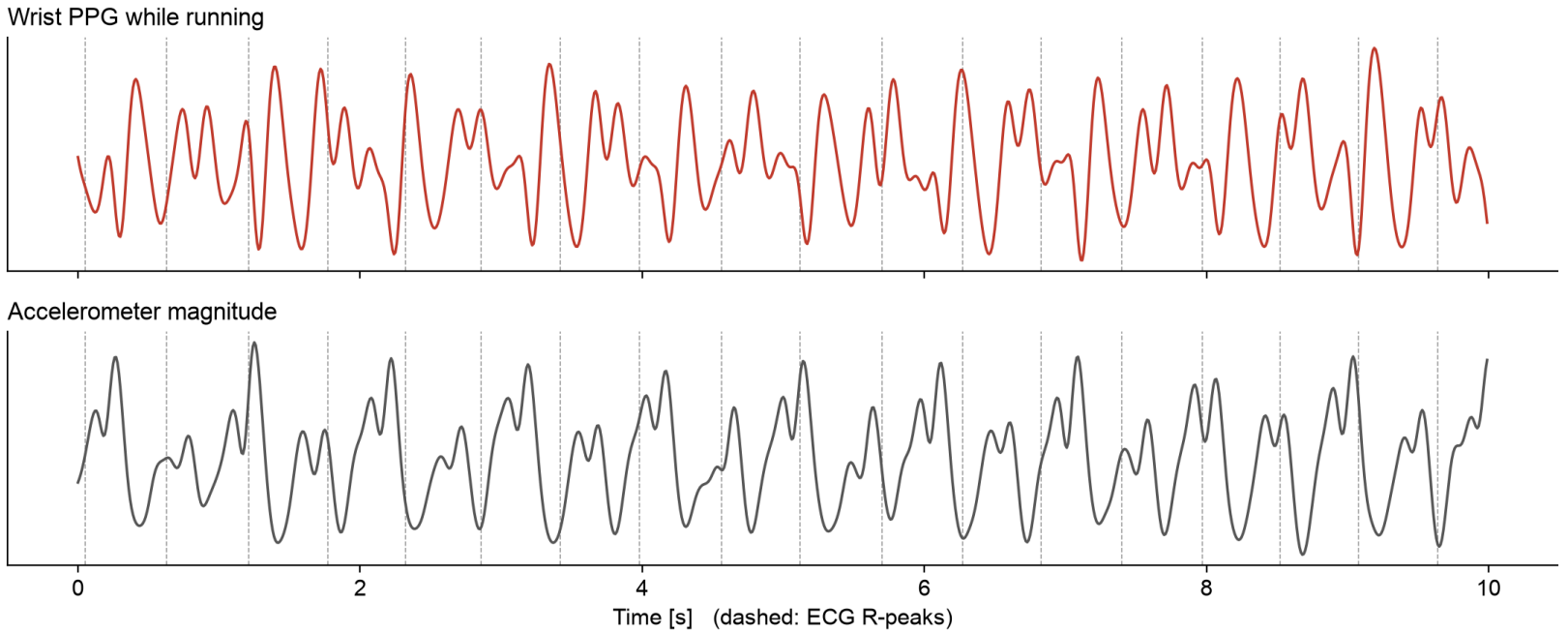


Filtering: keep the beat, drop the drift

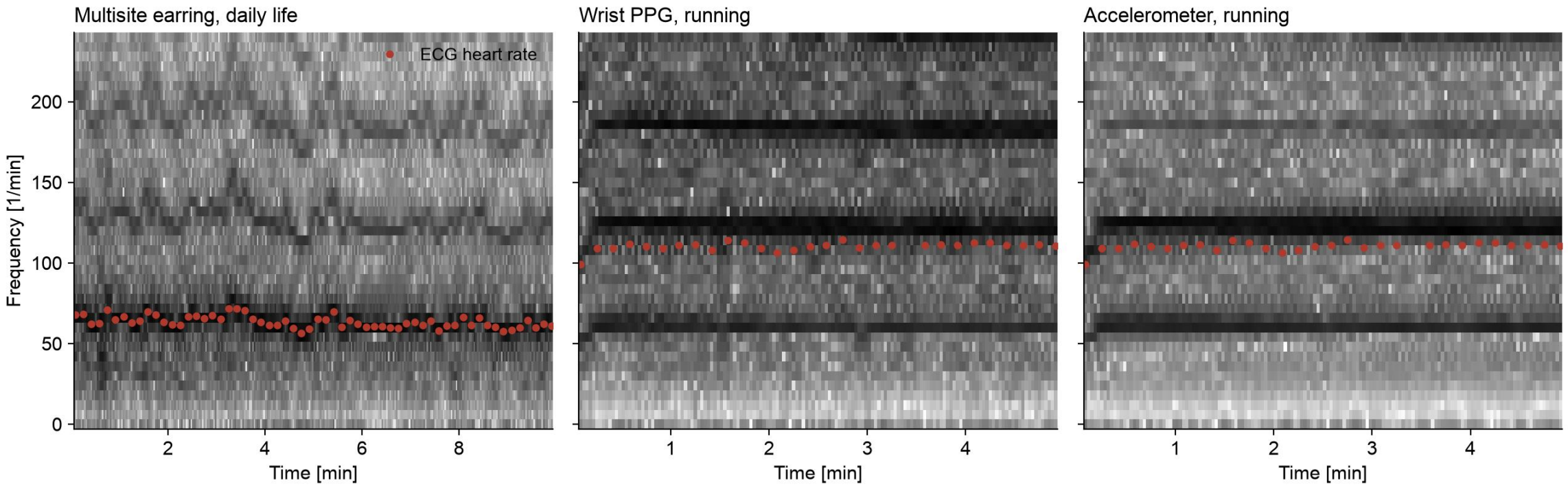


ECG 5–30 Hz: removes baseline wander and T-wave energy → sharp QRS for Pan-Tompkins. **PPG 0.5–8 Hz:** removes respiration/posture drift and sensor noise, keeps pulse and its harmonics.

Motion: the PPG follows the steps, not the heart



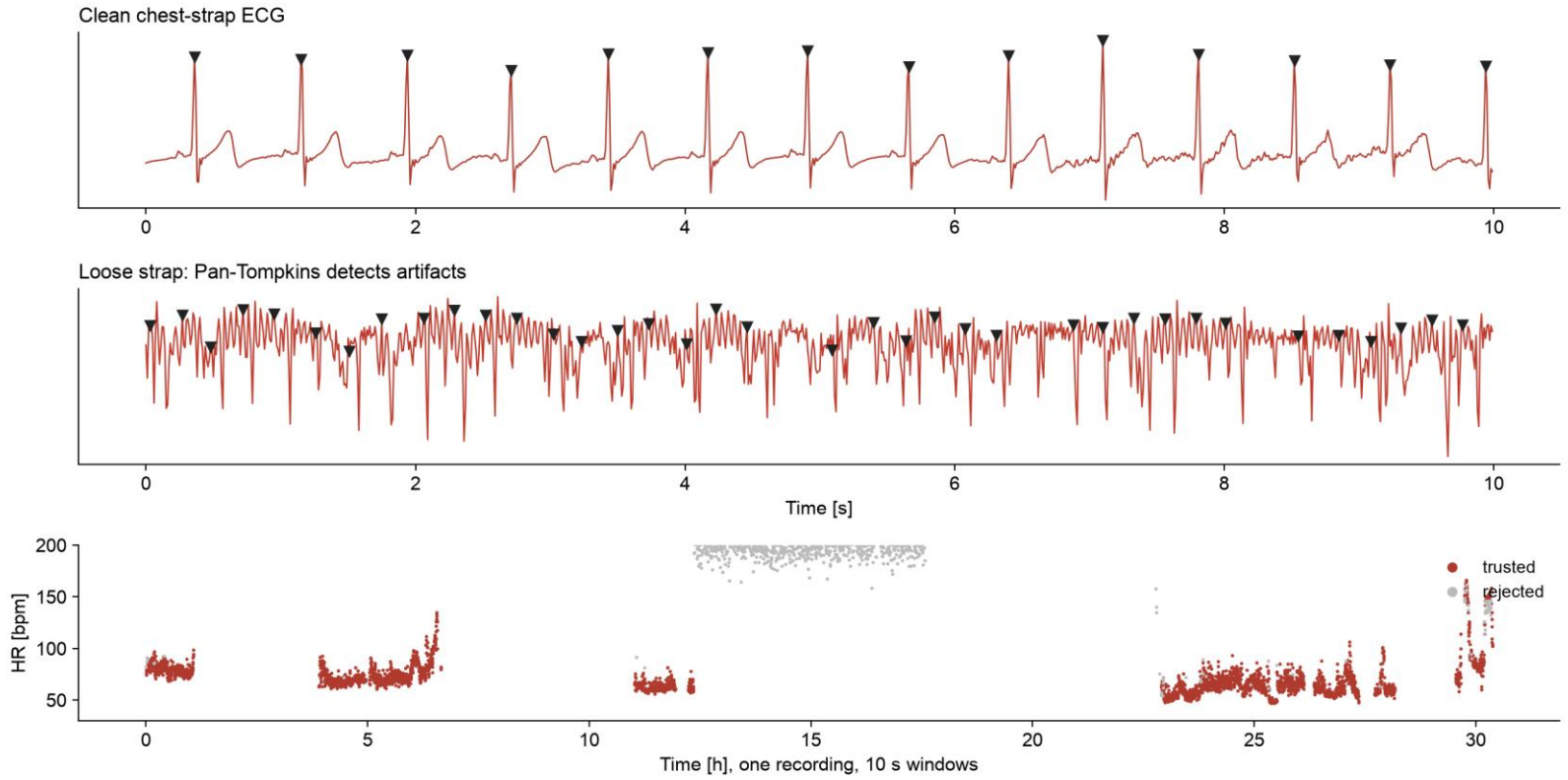
Time–frequency: heart-rate ridge vs. motion



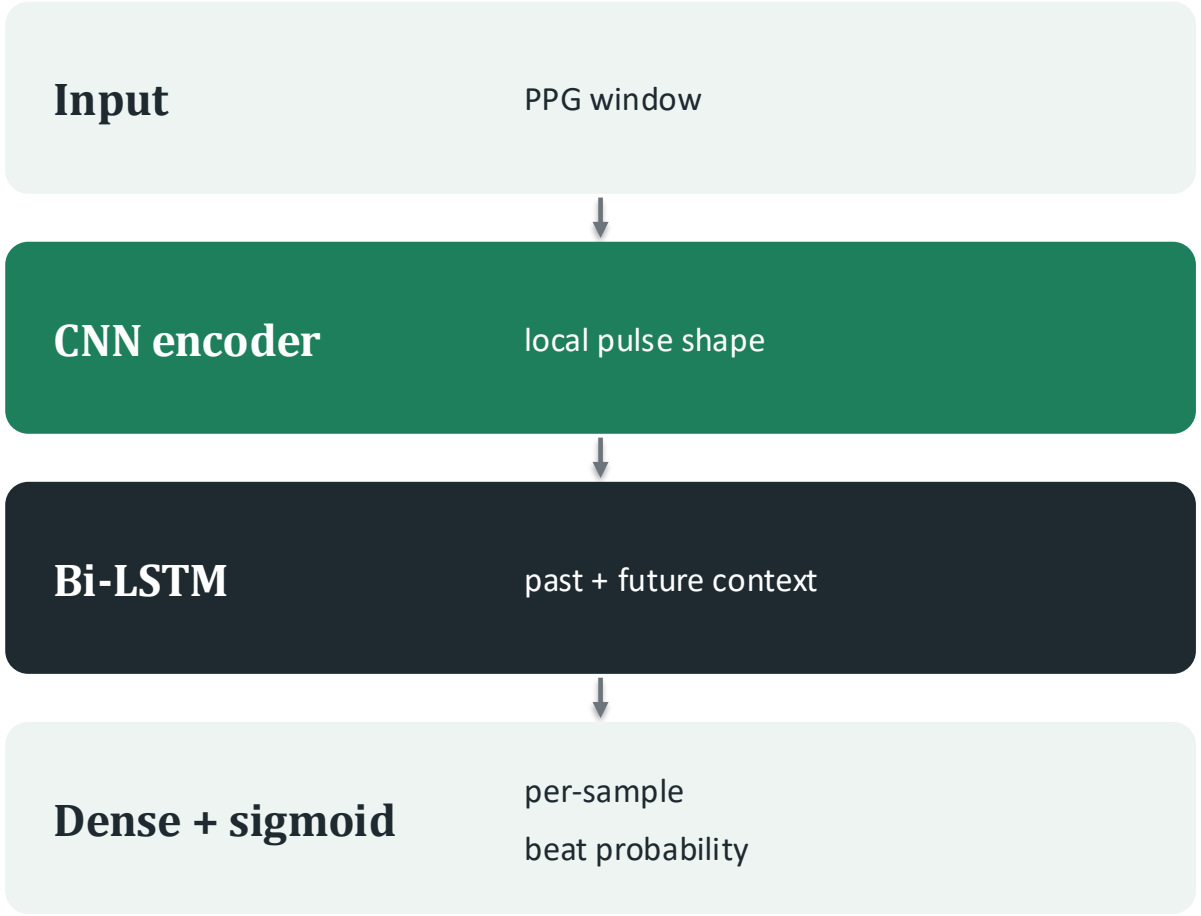
Daily life: PPG energy follows the ECG heart rate (red) → pulse recoverable.

Running: PPG energy sits at 62 / 125 / 187 per min, the arm-swing and step harmonics also seen in the accelerometer; the heart-rate ridge (~110 bpm) is buried.

Ground truth: Pan-Tompkins on the ECG



Our choice: own Bi-CNN-LSTM beat detector



CNN extracts morphology (upstroke, notch) and suppresses noise.

Bi-LSTM enforces rhythm consistency across beats; motion-corrupted beats can be inferred from neighbors